

Analiza rynku samochodowego w Polsce

Car Registration in Poland, 2021-2025 — Power BI Project Case Study

Data engineering · dimensional modeling · DAX · interactive reporting

Executive summary

Designed and built an end-to-end Power BI report analyzing 5 years of Polish vehicle registration data (2021-2025, 53,151 rows across 60 source files), covering data engineering, dimensional modeling, DAX and interactive report design.

Highlights

- Wrote a Python (pandas) script to clean and standardize 60 monthly open-data files — encoding fixes, Polish diacritics normalization, brand-name consolidation — then loaded and shaped the results in Power Query.
- Designed a star-schema data model: 2 fact tables (registrations, population) and 4 dimension tables, plus a 16-measure DAX layer covering YoY / MoM time intelligence and population-normalized KPIs.
- Implemented advanced Power BI features: a field parameter (dynamic month/quarter switch), 2 custom report-page tooltips, conditional formatting (icon sets and a colour-scale choropleth map), and slicers synchronized across pages.
- Ran a self-review pass after initial delivery and fixed 3 measure/visual defects found during QA: a misleading yearly total, a false previous-year comparison, and an uncoloured map.

Skills demonstrated

Area	Skills / tools
Data engineering & ETL	Power Query (M), Python, pandas, data cleaning, encoding and locale handling, deduplication
Data modeling	Star schema design, fact/dimension tables, calculated columns, cardinality and relationship design
Analytics (DAX)	Time intelligence (YoY, MoM), CALCULATE, population-normalized KPIs, field parameters
Report design	KPI cards, choropleth maps, conditional formatting, custom tooltip pages, cross-page filter sync, accessible colour use
Business analysis	Requirements-to-model mapping (course brief → data model), stakeholder-ready storytelling, QA and defect triage

Project overview

This project is an interactive Power BI report, "Analiza rynku samochodowego w Polsce" (Analysis of the car market in Poland), covering passenger car registrations in Poland from January 2021 to December 2025. It combines 60 monthly open-data files with population data, so the market can be viewed by year, quarter and month, by voivodeship and macroregion, and by brand, category and producer continent. Every figure is compared with the previous year (YoY) or previous month (MoM), and registrations are also related to population (registrations per 1,000 residents).

The report was prepared for the course Business Intelligence Software (Poznań University of Economics and Business, Data Analytics in Business and Controlling), whose brief required a star-schema model, DAX measures, multi-page navigation with synchronized filters and advanced Power BI features. The next section shows how each requirement was met.

Business questions and course requirements

The report answers:

- How many cars were registered per year, quarter and month, and how does that compare with the previous year and previous month?
- Which brands, categories and producer continents dominate registrations?
- How strong is the German trio (Audi, BMW, Mercedes-Benz): its share of registrations and the change in percentage points?
- How do voivodeships and macroregions compare, also relative to their population (registrations per 1,000 residents)?

Course requirements and how they are covered:

Requirement (course brief)	How the project meets it
At least 1 fact table and 3 dimension tables	2 fact tables (registrations: 53,151 rows; population) and 4 dimensions (calendar, year, location, car).
At least 1 calculated column and 5 measures, including 2 advanced CALCULATE measures	Calculated columns such as "Województwo z kodem TERYT" and "Kwartał tekst"; 16 measures in the measure table, including CALCULATE-based time-intelligence measures (previous year/month) and CALCULATE-filtered measures for the German-trio share.
Sensible formatting, at least 1 custom format	Custom formats visible on the KPI cards: percentage points (" +0,15 pp") and millions ("1,43M"); percentages, thousands separators and K-scaling elsewhere.
At least 2 pages with synchronized filters	3 main pages. The Year slicer is synced across all 3 pages; the Voivodeship slicer is synced between Introduction and Brands.
At least 8 visuals	10 analytical visuals on the 3 main pages (3 KPI card groups, bar chart, area chart, line chart, filled map, 3 tables/matrices), plus 2 tooltip pages.
At least 3 advanced features	Conditional formatting (icons and colours for YoY/MoM), 2 report-page tooltips, a field parameter (Month/Quarter switch).
Aesthetics and readability	Consistent light-blue theme, custom page background with title banner, clear KPI hierarchy, PY shown in grey next to CY in blue, dynamic licence-plate card.

Data sources

Item	Details
Source	dane.gov.pl (monthly vehicle registration files) and stat.gov.pl (population), as credited in the report footer.
Coverage	60 monthly files, January 2021 - December 2025
Granularity	Month × voivodeship × brand (fact table: Marka, Liczba, Województwo, Data)
Size after cleaning	53,151 rows in the registrations fact table
Licence / terms	Public open-data sources (dane.gov.pl, stat.gov.pl).
Known limitations	Data starts in January 2021, so 2021 has no previous-year comparison. Brand category (6 groups) and producer continent (4 groups) come from a project-specific brand metadata table covering 160 brands.

Data cleaning and transformation

A two-stage pipeline prepared the 60 monthly CSV files: a Python (pandas) script for encoding and brand-name cleanup, followed by Power Query for shaping and loading into the model.

- Python (pandas): re-encoded all files as utf-8-sig for Polish locale compatibility, normalized Polish diacritics across the 16 voivodeships, and consolidated brand-name variants (e.g. MERCEDES-AMG → MERCEDES-BENZ, while deliberately keeping OMODA and JAECCO separate from CHERY).
- Python (pandas): filtered out non-car vehicles before the files reached Power BI.
- Power Query: loaded the cleaned monthly files, enriched them with brand metadata (160 brands with producer continent and category) and a location table with macroregion and TERYT code.

Marka	Liczba	Województwo	Data
LINCOLN	1	Pomorskie	styczeń 2021
LINCOLN	1	Podkarpackie	styczeń 2021
LINCOLN	1	Kujawsko-Pomorskie	styczeń 2021
LINCOLN	1	Dolnośląskie	styczeń 2021
LINCOLN	1	Mazowieckie	styczeń 2021
LINCOLN	1	Pomorskie	luty 2021
LINCOLN	1	Podkarpackie	luty 2021
LINCOLN	1	Lubuskie	luty 2021
LINCOLN	1	Lubelskie	luty 2021
LINCOLN	1	Wielkopolskie	luty 2021
LINCOLN	1	Lubelskie	marzec 2021
LINCOLN	1	Dolnośląskie	marzec 2021
LINCOLN	1	Zachodniopomorskie	marzec 2021

Figure 1. Registrations fact table after cleaning (Power BI Data view, 53,151 rows).

Data model

Star schema with two fact tables sharing conformed dimensions.



Figure 2. Data model (Model view): fact tables, dimensions, measure table and relationships.

Tables in the model

Table	Role	Contents
DANE O REJESTRACJACH	Fact	Data, Liczba, Marka, Województwo (columns hidden from the report view)
LUDNOŚĆ	Fact	Liczba ludności, Rok, Województwo (population by year)
KALENDARZ	Dimension	Data, Rok, Kwartał, Miesiąc, Nazwa miesiąca, Kwartał tekst (calc.), Rok-Miesiąc tekst
WYMIAR ROK	Dimension	Rok: year dimension behind the Year slicer, linking calendar and population
LOKALIZACJA	Dimension	Województwo, Makroregion, Kod TERYT, Województwo z kodem TERYT (calc.), Wyróżnik tablic (licence-plate prefix)
SAMOCHODY	Dimension	Marka, Kategoria, Kontynent
MIARY	Measures	16 DAX measures (see section 6)
[Kontekst] Skala czasu	Field parameter	Switches the trend chart between Month and Quarter
[Kontekst] Miary na wykres	Helper measures	Registrations per 1,000 residents (CY and PY) and licence-plate prefix, used in tooltips and cards

Relationships: KALENDARZ, SAMOCHODY and LOKALIZACJA each filter DANE O REJESTRACJACH (one-to-many). LOKALIZACJA filters LUDNOŚĆ (one-to-many). WYMIAR ROK connects KALENDARZ and LUDNOŚĆ (one-to-many, both filter directions).

Key DAX measures

The measure table (MIARY) holds 16 DAX measures, organised around time comparison, population normalisation and the German-trio analysis:

- Liczba zarejestrowanych samochodów — registrations in the selected period
- ...PY / ...YoY % — same period of the previous year, and the percentage change
- ...PM / ...MoM % — previous month, and the month-over-month change
- Liczba ludności / ...PY / ...YoY % — population, previous-year population and its change
- ...na 1000 osób / ...PY / ...YoY % — registrations per 1,000 residents, previous year and change
- Liczba...z niemieckiej trójcy — registrations of Audi, BMW and Mercedes-Benz
- Udział niemieckiej trójcy... / Zmiana udziału...(pp) — the trio's share of all registrations, and its change in percentage points vs the previous year
- Suma wszystkich rejestracji samochodów — registrations across all years (ignores the year slicer)

The previous-year, previous-month and "all years" measures use CALCULATE to override the year/month filter context; the German-trio measures use CALCULATE with a brand filter.

Report walkthrough

Page 1: Introduction (Wprowadzenie)

Year and Voivodeship slicers, KPI cards for registrations in the selected year (CY), the previous year (PY) and all years, and a block for the German trio: registrations, market share and change in percentage points. A licence-plate card shows the plate prefix of the selected voivodeship.

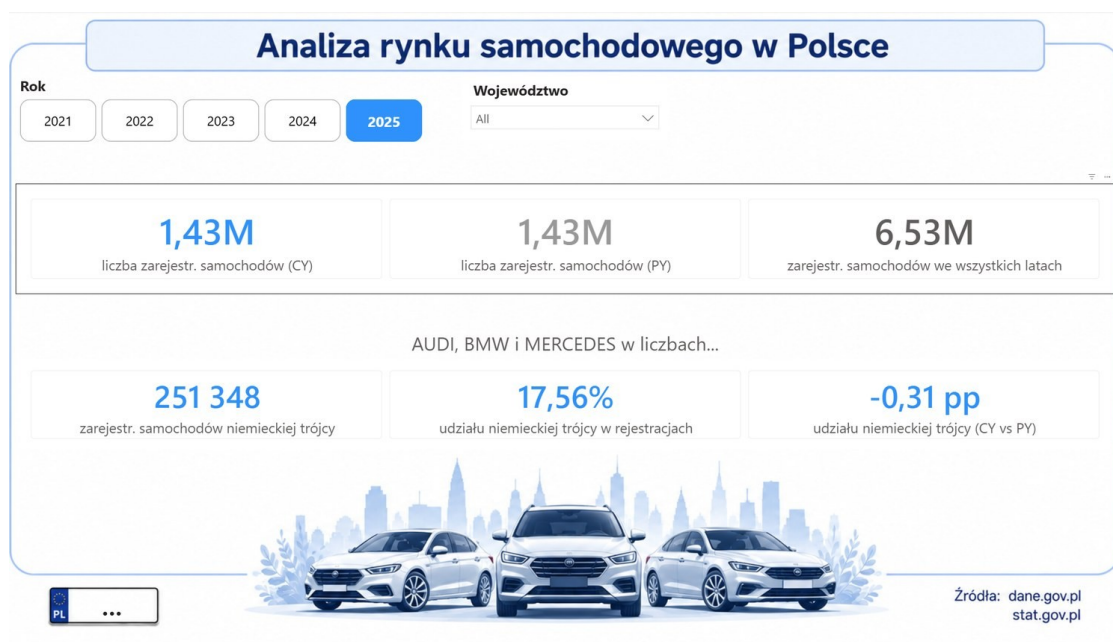


Figure 3. Introduction page, all of Poland, 2025.

Page 2: Brands (Marki)

Top 10 brands (bar chart), registrations per month with previous month and MoM % (table with icons), registrations by category (area chart, CY vs PY) and by producer continent (table with YoY icons). The "Cały rok" (full-year) row shows the annual total only; the PM and MoM columns are left blank there, since a sum of shifted monthly comparisons is not a meaningful year-over-year figure.



Figure 4. Brands page, all of Poland, 2025 — full-year row now shows only the total.

Page 3: Registration map (Mapa rejestracji)

A Month/Quarter switch (field parameter) drives the line chart of registrations against the previous year. The map is colour-scaled by registrations per 1,000 residents (darker = more), with a subtitle explaining the scale. Below: a matrix of macroregions and voivodeships with population, registrations and YoY icons.

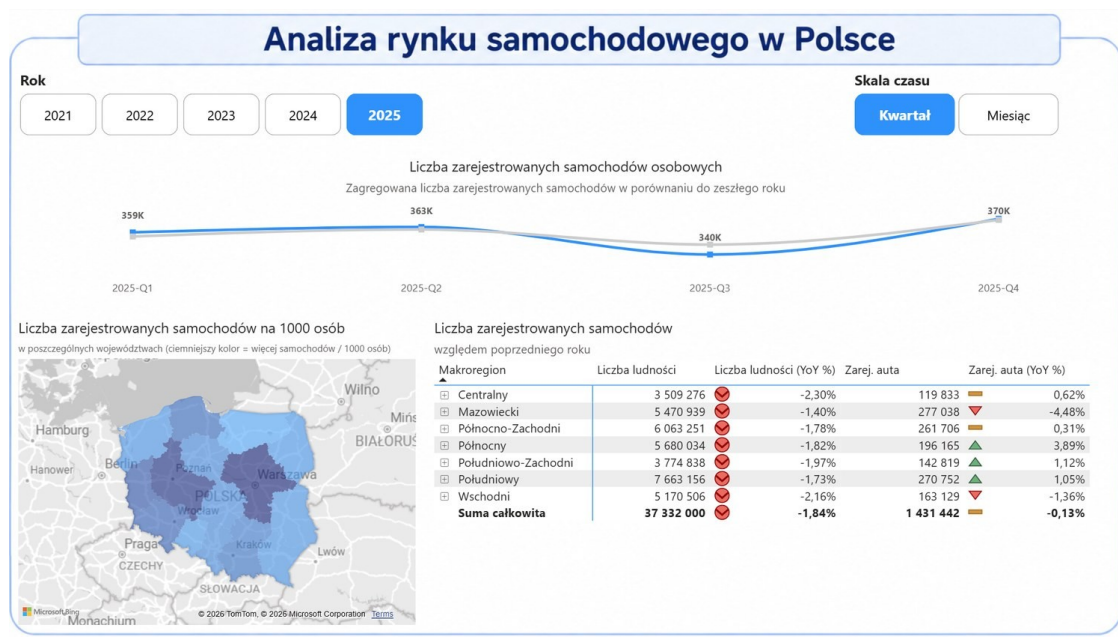


Figure 5. Registration map page, quarterly view, 2025 — map coloured by registrations per 1,000 residents.

The same chart can also be switched to monthly granularity via the Skala czasu field parameter, without changing anything else on the page:



Figure 6. Trend chart switched to monthly view (Miesiąc) for the same 2025 data.

Filtered to a single voivodeship — Wielkopolskie, 2025

The Year and Voivodeship slicers filter all three main pages at once. Selecting a single voivodeship still returns a meaningful, complete report: KPIs, brand ranking and the map all update consistently.

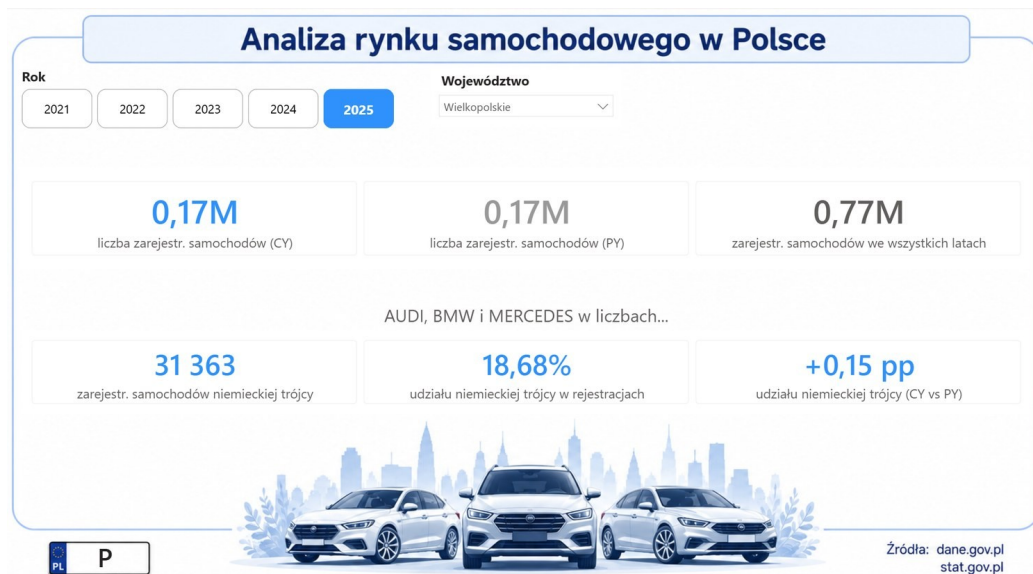


Figure 7. Introduction page filtered to Wielkopolskie, 2025.

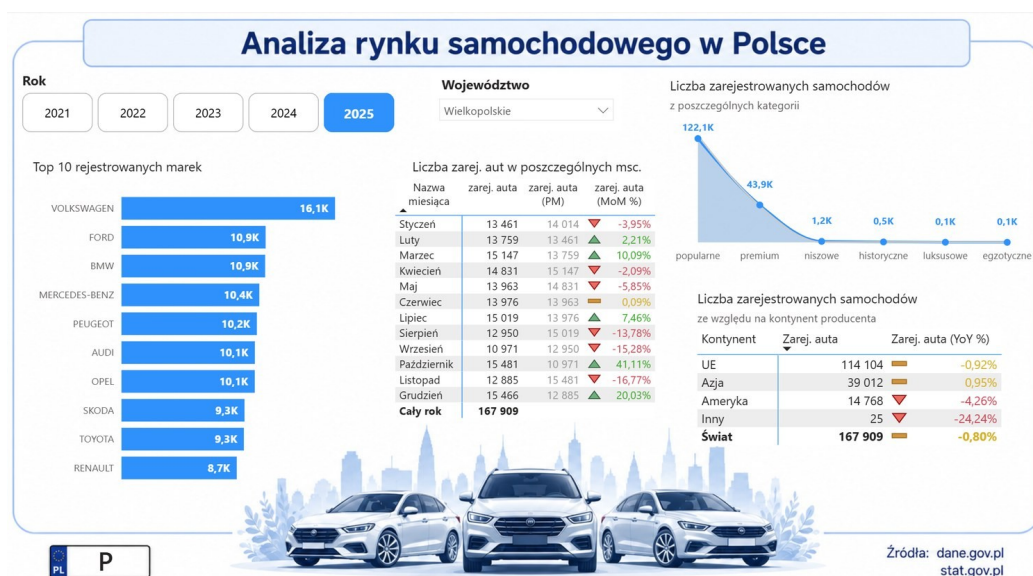


Figure 8. Brands page filtered to Wielkopolskie, 2025.

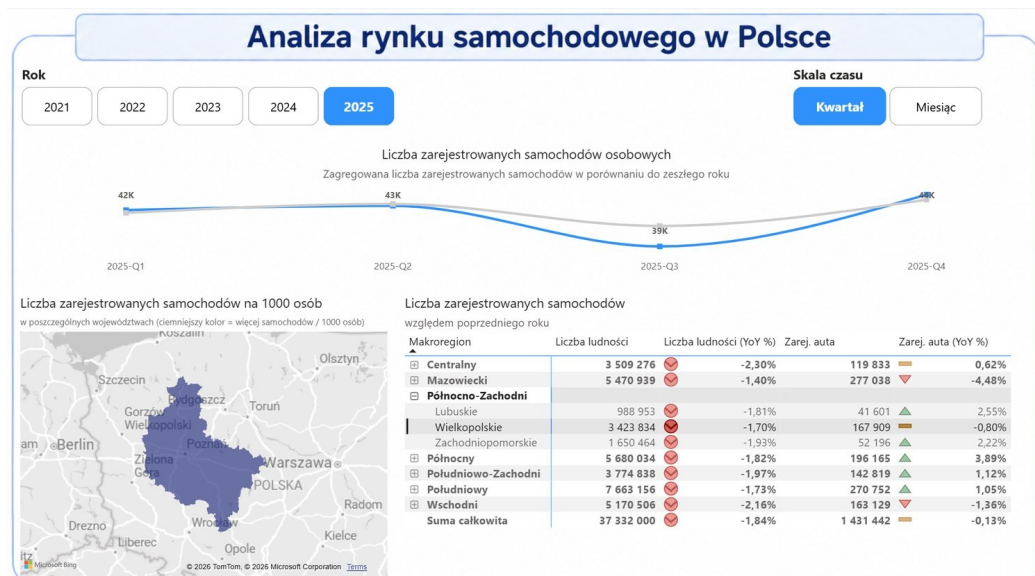


Figure 9. Registration map page, quarterly view, Wielkopolskie highlighted.

Edge case: first year of data

2021 is the first year in the dataset, so there is no 2020 to compare against — this applies to every voivodeship in 2021, not just the one shown below. Rather than silently showing a misleading number, the previous-year and percentage-point measures are written to return "--" whenever the comparison period doesn't exist. The example below filters to Pomorskie to show this on the PY and CY-vs-PY cards, but the same handling applies across the board. This was a deliberate defensive-DAX decision, not a display error.

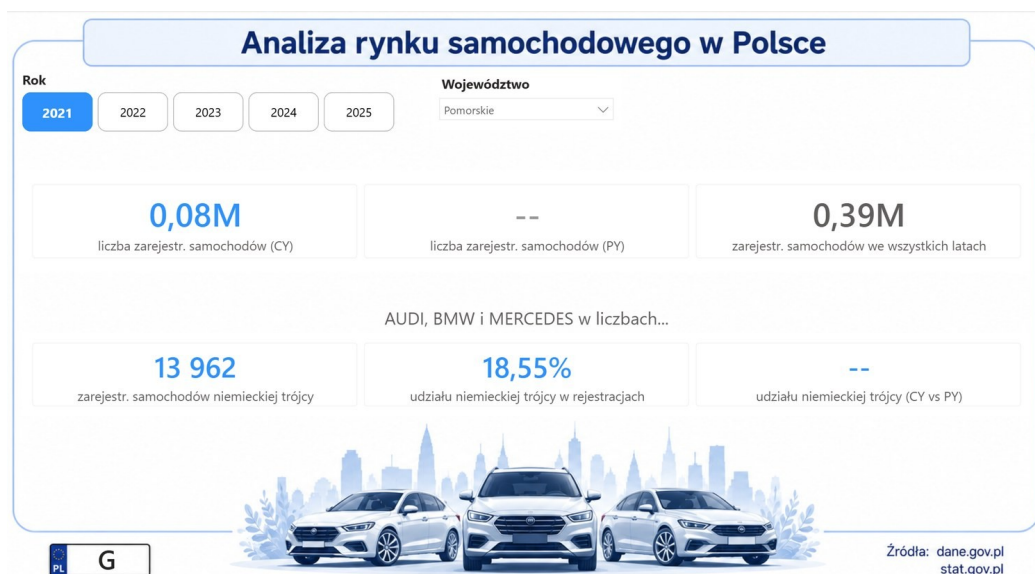


Figure 10. Introduction page filtered to Pomorskie, 2021 (used here as one example) — PY and change cards correctly show "--", not a false value.

Tooltip pages

Two custom report-page tooltips give extra detail on hover: one for the map (registrations per 1,000 residents, CY vs PY) and one for the population table (previous-year population and registrations per 1,000 residents).

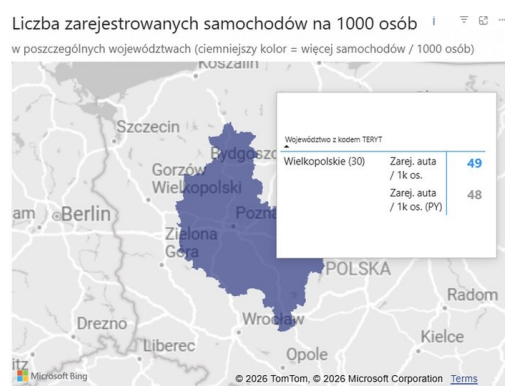


Figure 11. Tooltip page "Tooltip Województwo": registrations per 1,000 residents, CY vs PY.



Figure 12. Tooltip page "Tooltip l.s. PY": previous-year population and registrations per 1,000 residents.

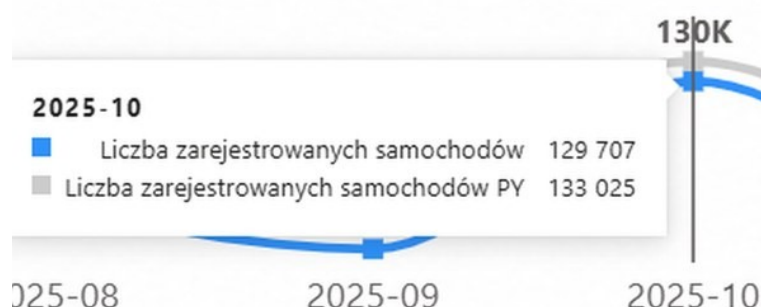


Figure 13. Default hover tooltip on the trend chart: current vs previous year for a single month (October 2025).

Findings (2025)

1. **Flat market.** 1,431,442 cars were registered in 2025, -0.13% vs 2024; 6.53M across 2021-2025.
2. **Seasonality.** Monthly registrations range from 101.7K (September) to 131.7K (December); March (130.7K) is the other peak. October rebounds +27.49% on September.
3. **Mass market dominates.** The popular category accounts for 1,044.3K registrations (about 73%), premium for 373.0K (about 26%); niche, historic, luxury and exotic together are under 15K.
4. **Brand ranking.** Volkswagen (120.6K), Toyota (117.2K), Audi (93.6K), Ford (92.3K) and Skoda (89.0K) lead the top 10.
5. **German trio.** 251,348 registrations, a 17.56% share (-0.31 pp vs 2024). In Wielkopolskie: 31,363 registrations, 18.68% share (+0.15 pp).
6. **Producer continent.** EU-made cars 869,852 (-1.01%), Asia 427,498 (+1.63%), America 133,862 (+0.10%).
7. **Regions.** Mazowiecki fell most (277,038, -4.48%), Północny grew most (196,165, +3.89%). Wielkopolskie registered 167,909 cars (-0.80%).

Challenges and lessons learned

The main challenge was data quality across 60 files: inconsistent encodings, Polish diacritics, brand names that differ between months, and non-car vehicles mixed into the registrations. A Python (pandas) script handled that cleanup before the files reached Power Query, which made the model reliable. The model also keeps a separate population fact table and year dimension to support the previous-year and per-1,000-residents measures.

A second lesson came from reviewing the finished report against its own logic rather than just its visuals: a seemingly correct yearly total (Brands page) and a seemingly correct percentage-point card (2021) both turned out to mask edge-case DAX behaviour — a total row silently summing mismatched previous-month values, and a missing-previous-year measure defaulting to zero instead of blank. Both were fixed after review. It reinforced the habit of checking totals and edge periods (first year, filtered views) before publishing, not just the headline numbers.

Roadmap

Planned next steps: add fuel type / electric-vehicle share, extend the model with a forecast of monthly registrations, automate the monthly refresh of the source files, and finish validating the population series against a single consistent source.